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Shimmer 3 GSR+: Sensor Placement and Respondent Instructions

This article describes the theory behind optimal GSR collection, and guidelines for sensor placement.

Measurement of Galvanic Skin Response is accomplished as exosomatic DC (i.e., noninvasive direct current) recording from an electrode pair (bipolar recording). A low constant voltage is applied, and the voltage difference between the two electrodes is then reported as GSR signal of the bipolar channel.

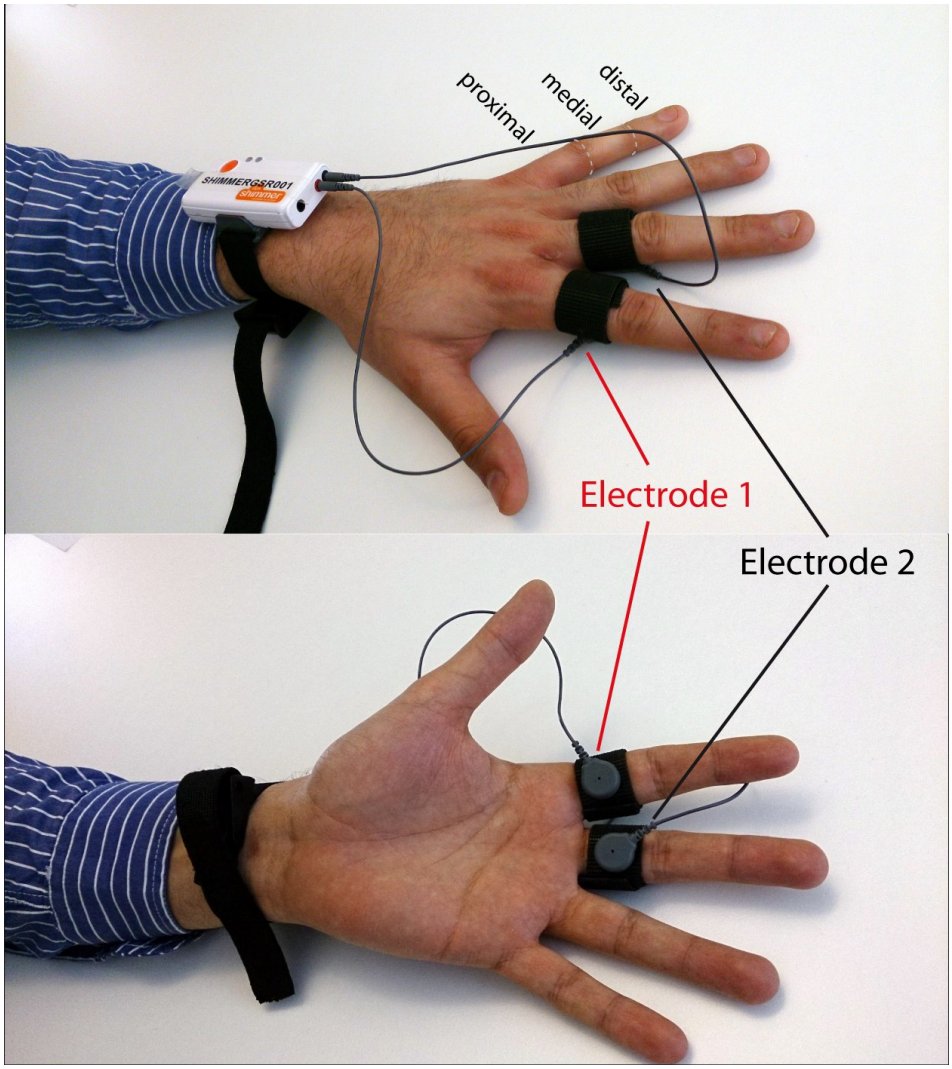
The GSR data collected in iMotions represents skin conductance (SC), which has the unit siemens (S) or mho (units are equivalent). Since skin conductance values are quite low, the data is expressed in micro-Siemens (µS).

iMotions also provides GSR as a function of resistance in kilo-Ohms (kOhm). Resistance is the inverse function of conductance, thus the graphs in µSiemens and kOhms are mirror images of one another. However, analysis is conventionally done on µSiemen values.

1. Recording Sites

The GSR electrodes are placed on the volar phalanges of the fingers (proximal, medial or distal) on the palms of the non-dominant hand. The distal phalanges of the fingers are preferred because of their greater responsivity (Scerbo, Freedman, Raine, Dawson, & Venables, 1992). If participants have to use both of their hands during EDA recording, the next best recording sites would be the plantar ones (feet). As the soles are normally subjected to pressure during standing or walking, Edelberg (1967) recommended a medial site on the inner side of the foot instead, over the abductor hallucis muscle (the extensor of the big toe's base joint) adjacent to the foot sole, and midway between the proximal phalanx of the big toe and a point directly beneath the ankle.

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Note: Electrode 1-2 are interchangeable

The Optical Heart Rate sensor is placed either on another free finger (Shimmer Optical Pulse Probe sensor) or applied to the earlobe (Shimmer3 Ear Clip sensor, 2015 and later).

2. Pretreatment of Recording Sites

Using the Shimmer GSR+ sensor, no special pretreatment is required. However, if the participant has very dry hands, hand cream should be used to moisturize the skin.

3. Respondent Instructions

- Instruct respondent to find a comfortable position and to avoid any unnecessary movement during the experiment, especially those of the limbs from which GSR is recorded.
- Also, instruct respondents to breath normally, they should not hold their breath or breathe irregularly (record the respiration curve in addition to EDA recording to enable a later elimination of artifacts caused by

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irregular respiratory activity).

- Tell subject that talking should be avoided during electrodermal recording whenever possible.
- It might be a good idea to show respondents in the iMotions live viewer how movements and breathing affect the signal: While movements result in high-frequency spikes that distort the GSR signal, excessive breathing or holding of one’s breath results in slow drifts in the GSR data that could be misinterpreted as punctual bursts reflecting increased arousal of the respondent.

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